The Domestic Political Economy of Net Zero

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1 Introduction

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2 Analytical primitives

To elucidate the comparative political economy (CPE) of climate policy, some analytical groundwork is necessary related to the framework's primitives. Doing so also gives us the opportunity to synthesise much of the relevant comparative political economy and climate politics literatures along the way. The framework's analytical starting point is familiar for (comparative) political economists (Hacker and Pierson, 2014; Arnold, 1990). Democratic policymakers navigate electoral pressure from voters and organised-interest pressure from actors whose resources, information, investment decisions, and cooperation in implementation affect their ability to win office and enact policy. The CPE toolkit is well suited to analysing each pressure, though it has less to say about how the two interact, especially when policy adopted in one period changes the electoral and interest-group constituencies as well as expectations that structure politics in the next.

Climate policy is a distinctive domain for studying this interaction for three reasons. First, mitigation is a long-term policy problem in the sense developed by Jacobs (2011b, 2016) and Lindvall (2010, 2017). That is, governments must impose near-term costs on voters and producers in the pursuit of benefits that lie not only in the future but are more uncertain and diffuse than the costs of these

policies.[1] Climate policy therefore requires governments to allocate costs now for benefits that are delayed, uncertain, and politically difficult to claim credit for—both because they materialise only in the future and take the counterfactual form of avoided climate damages rather than tangible benefits. The second characteristic is the intermittent salience of climate policy. Climate change matters to many voters, but it is rarely the issue they attach most important to or the main axis of political competition for sustained periods (Egan and Mullin, 2017; Kenny, 2022). Third, for fossil-fuel producers and carbon-intensive sectors, stringent mitigation policy entails the devaluing of their assets—typically referred to as the risk of stranded assets (van der Ploeg, 2016; von Dulong et al., 2023)—because it reduces the future economic value of assets tied to activities that deep decarbonisation aims to phase out. Indeed, when dirty producers have few viable means of substituting to cleaner methods of production, climate policy becomes an existential risk for them (Colgan et al., 2021).

These characteristics make climate policy a distinctive long-term policymaking problem. Like pensions, infrastructure, or education, climate mitigation requires governments to impose costs now for benefits later. What sets climate apart is that this intertemporal trade-off operates in a setting where electoral attention is intermittent and incumbent producers face losses in the value of assets on which their business models depend. This means that climate policy does more than allocate costs across actors and time. It also changes the political conditions under which later policy is made. Electoral and producer pressures determine policymakers' incentives in the present, while today's policy choices can reshape the electoral and producer environment in which tomorrow's choices are made.

This structure determines the framework's dependent variable. The outcome of interest is the *durability of climate-policy stringency*. Stringency refers to the ambition of climate policy and durability to its persistence. Durability matters because ambitious enactment and durable policy change can diverge. A stringent but easily repealed carbon price, a legally binding target without instruments, and a policy that survives turnover and changes behaviour would otherwise be treated as equivalent. We therefore abstract from symbolic targets not backed by instruments and treat emissions reductions as downstream outcomes. Since durability varies across sectors and instruments within countries, the relevant unit of analysis is the policy

[1] The combination of diffuse benefits and concentrated costs is often referred to as an Olsonian distributional profile (Edenhofer and Flachsland, 2025).

episode.

2.1 The electoral side

On the electoral side, the framework's explanatory variable is the *state of electoral demand*. We do not seek to explain voters' preferred level of climate-policy stringency or its determinants.[2] Instead, we are interested in the question as to whether, and in what way, those preferences enter the policymaker's electoral calculus. Two factors determine the answer, we argue. The first is salience, meaning whether climate is important enough relative to other issues to affect political behaviour and competition. The second is the clarity of the electoral signal, meaning whether, once climate is salient, the distribution of preferences gives the policymaker an identifiable vote-maximising direction.[3] Together they generate three types of electoral demand: latent demand as well as noisy or clear activation.

Before expanding on these, let us clarify our assumptions about the policymaker. We assume the policymaker to be office-seeking and myopic, focused on winning the next election. This is not because we deem policy-seeking motivations—whereby ideology, personal conviction, or the preferences of one's core electorate drive policy decisions (Strom, 1990)—irrelevant.[4] Instead, our focus on office-seeking behaviour serves an analytical purpose. By holding politicians' motivations and the selection of politicians constant, the assumption isolates the climate-policy incentives generated by the need to win office again and placate producers. In doing

[2]That said, any account of electoral demand presupposes some conceptualisation of voter preferences. Given the many surveys of the literature on climate preferences, including Mattauch et al. (2026) in the present issue (see also Fairbrother (2022); Gazmararian et al. (2025)), we prescind from expanding on it here (see e.g. Gaikwad et al. (2022); Bayer and Genovese (2025)). Suffice it to say, the central assumption underpinning our analysis is that voters' demand for climate policy declines as the material costs they have to bear increase. This is consistent with the finding that support for climate policy is broad (Andre et al., 2024) but shallow, i.e. highly cost-sensitive (Dechezleprêtre et al., 2025). Costs become electorally consequential when they affect voters, especially in ways that are clearly attributable to government policy rather than to markets or external shocks (Arnold, 1990; Gazmararian et al., 2026).

[3]The "median voter" is often used as a shorthand for the vote-maximising position even when it does not literally refer to the population median. We assume only that some such position exists; under simple-plurality-single-member-district systems, for instance, the relevant position is whatever secures pivotal voters in marginal constituencies (Wiedemann, 2024), rather than the population median.

[4]Indeed, we follow Lindvall et al. (2023) in arguing that policy-seeking motivations loom larger when the electoral signal is weak or unclear. They argue that when the median voter's position is clear, office-seeking motivations dominate; when it is uncertain, core voters become the electoral anchor. We revisit this argument below (XXXX).

so, we deliberately stack the deck against long-term policymaking, which allows us to probe the robustness of climate incentives under representative democracy.

Latent demand. Demand is latent when voters have climate preferences, but climate does not structure political behaviour or competition. Voters care about many issues, weight them unequally, and often—as noted above—accord climate less priority than other issues like prices, employment, security, health, or immigration (Egan and Mullin, 2017; Gazmararian et al., 2025). This implies that the benefits of climate action engender very limited electoral benefits. Conversely, inaction generates limited electoral costs, even if voters support climate policy in principle. The result is what V. O. Key (1961) called latent opinion: preferences that could become politically consequential but are not currently driving behaviour.

The implication of latency is an asymmetry in the policymaker's electoral incentives. The benefits of climate action matter less to voter than its costs, especially when these costs are traceable—i.e. attributable to or perceived as being so—to government policy (Arnold, 1990). Inaction is therefore electorally attractive since it entails no (additional) costs for voters. When policymakers do act, however, they have incentives to hide or diffuse costs (Mildenberger and Sahn, 2025).[5] The choice and design of instruments then become central, with instruments that render costs opaque for voter being particularly popular. Latent electoral demand thus logic constrains the design of action, but does not determine whether action occurs; it allows for both action and inaction. Explaining which one course of action is pursued requires an analysis of producer pressure, which we develop below (Section XXXX).

Activation: salience and clarity. Climate preferences become activated when climate becomes electorally relevant enough to affect political behaviour and competition. Much of the literature focuses on exogenous drivers, such as natural disasters[6] (Hoffmann et al., 2022; Hilbig and Riaz, 2024; Baccini and Leemann, 2021; Garside

[5] As Arnold (1990) notes, there are two central margins of cost diffusion. Temporal diffusion pushes costs into the future or spreads them over time. Geographic diffusion spreads costs across regions so that no single constituency bears a concentrated burden, which is especially relevant in countries with single-member plurality districts, where the spatial distribution of preferences matters for electoral success (Rodden, 2019).

[6] It is worth noting, however, that this burgeoning literature on natural disasters finds at best short-lived effects of disasters on the importance individuals attach to climate or on pro-climate voting behaviour.

and Zhai, 2022) or energy price shocks (Konc et al., 2024). But the salience of climate policy can also change endogenously. Policymakers and parties shape it through elite cues and issue emphasis (Merkley and Stecula, 2021; Carmichael and Brulle, 2017; Wagner and Meyer, 2014; Sio and Weber, 2014; Schwörer, 2024; Sanford et al., 2025); social movements through protest (Gause, 2022; Brehm and Gruhl, 2024; Valentim, 2025); and interest groups through framing and the mobilisation of aligned electoral constituencies (Brulle and Werthman, 2021). We use "activation" rather than salience because activation captures the movement from latent preferences to electoral relevance, regardless of whether that movement is driven by exogenous or endogenous factors.

Clear activation. Salience, however, is not enough to determine the electoral incentives of the policymaker. For that, we need to know the underlying distribution of preferences, which determines the clarity of the electoral signal. Electoral signals are clear when activated preferences are sufficiently tightly clustered so as to enable the policymaker to identify a vote-maximising position, whether toward more or less climate action.[7] While often conflated, salience and clarity therefore differ (Busemeyer et al., 2020). Climate can be salient without producing a clear signal, and the underlying preferences can be concentrated in a particular direction without being activated, i.e. they can be latent. Clear activation also tends to make policy more durable than activation of the noisy kind as successor governments are more likely to face the same electoral pressures as its predecessor.

Even holding the degree of activation constant, however, electoral signals can become noisy when differential long-term value change in the population (Besley and Persson, 2023) increases the dispersion of the underlying distribution of preferences or when it becomes easier for anti-climate interest groups or parties to frame climate policy as a luxury good (Abou-Chadi and Kayser, 2017). When such appeals resonate, voters attach more importance to the costs of climate policy than its benefits. This is particularly likely during times when the budget constraints of governments or individuals tighten or when the costs of climate policy crystal-

[7] To ensure that such a position exists, assuming a uni-dimensional policy space is, if not strictly necessary, very convenient. But for "latent demand" to be meaningful at least two dimensions are required. To square what might seem like a circle, note that latency can be be reconciled with uni-dimensionality by assuming what we might call sequential uni-dimensionality, i.e. we assume that the policy space in any given period is uni-dimensional but this dimension is only sometimes climate policy.

lise for voters. We will return to this observation in our discussion of the future challenges of reaching net-zero targets (Section XXX).

Noisy activation. Activation is noisy when climate is salient but the electoral signal is unclear. This can happen because preferences are widely dispersed, because they are polarised (i.e. follow a bimodal distribution), or because competition between interest groups creates uncertainty about the true distribution. Without a clearly identifiable vote-maximising position, the office-seeking calculus alone does not pin down what the policymaker does.

Some additional anchor is required—three of which figure prominently in the literature. First, policy-seeking motivations loom larger, and policymakers fall back on the preferences of their core voters (Lindvall et al., 2023). Second, interest groups can supply the structure when they provide electoral resources that office-seekers value, such as campaign finance or the mobilisation of certain voter segments. Third, where feasible, policymakers can try to shift attention to non-climate issues, dampening climate's salience and avoiding the noisy domain altogether. Endogenous activation typically constrains this third option. The actors who raised salience have an interest in keeping it raised. The same competition also produces competing frames, adding to the noise (Druckman, 2022; Chong and Druckman, 2007).

Conditional on the policymaker having an interest in acting on climate, durability depends on two mechanisms. The first is insulation, where policy is removed from electoral and interest-group pressures through delegation to courts, independent regulators, or supranational bodies.[8] The second is policy feedback, where new beneficiaries—on the voter or interest group side—form quickly enough to raise the political cost of reversal before successors take office.

The limits of electoral demand. The above shows the potency and limits of electoral demand. Under clear activation, public opinion determines the policymaker's incentives for climate (in)action. But where electoral demand does not pin down policy choice—under latent demand, and sometimes under noisy activation—the

[8]The literature on delegation as a commitment device often treats institutional insulation as automatically solving the durability problem. This is too simple. First, voters may punish delegation itself when they perceive it as a loss of democratic accountability. Second, delegation is only stable as long as the political costs of maintaining it remain tolerable for the policymaker. Where electoral punishment or sanctioning by interest groups becomes severe, policymakers can and do reverse delegation.

framework needs another source of influence. Some accounts rely, as noted above, on core voters (Lindvall et al., 2023), policymakers' personal convictions, experiences, or socio-economic backgrounds; others on bureaucrats.

We zero in on what we regard as the empirically most important source: producers. The natural question is then why an office-seeking policymaker should care about what producers want, given that producers do not, as producers, vote. The answer is that producers affect the policymaker's ability to deliver what voters want. Two of the most prominent channels are electoral resources, such as campaign donations (Hall and Deardorff, 2006; You, 2022; Brulle and Downie, 2022), and the information necessary to implement smoothly what is electorally optimal (Minaudier, 2022).

2.2 The producer side

2.2.1 Two types of producers and their interests

We follow the political economy literature in dividing the producer side into two broad categories (Meckling, 2011; Kelsey, 2018; Genovese and Tvinnereim, 2019; Besley and Persson, 2023). The dirty sector consists of fossil-fuel-intensive firms. The clean sector consists of firms producing clean substitutes, like renewable electricity or electric vehicles. This dichotomous sectoral classification is a simplification that we adopt for the sake of analytical tractability. Clean firms have interests that mostly[9] align with climate action given that climate policy expands their market, raises dirty rivals' costs, and channels subsidies, procurement, and infrastructure investment toward them. Dirty firms, by contrast, lose from climate policy, ranging from compliance burdens and asset devaluation to existential risk when their business model is technologically incompatible with decarbonisation.

In what follows, we primarily focus on the dirty sector, with the clean sector entering principally as a counterweight to its dirty counterpart. This means we ignore the inner workings of clean-sector lobbying and influence—an important topic that is beyond the scope of our review. Within the dirty sector, the principal explanatory variable is whether dirty producers are unified or divided.[10]

[9] Clean firms differ in their stakes in increasing the stringency of climate policy.

[10] The obvious objection is that division or fragmentation may simply be an artefact of aggregation: individual dirty sectors may be united but disagree among them, leading to overall division. Our response is twofold. First, this is the price of simplification. Second, intra-sectoral heterogeneity

2.2.2 The firm-level calculus

Firm behaviour relating to climate policy is nuanced—ranging outright resistance to hedging[11] to active support (Vormedal, 2011; Meckling, 2015; Green et al., 2022)—but the actions discussed in the literature can be summarised along a continuum between *resisting* climate policy and *adjusting* to it. The location that a firm occupies on this continuum is determined by its adjustment costs, to wit the costs of converting carbon-intensive activities into low-carbon alternatives, or of repositioning away from carbon-intensive production altogether.[12] Firms with low adjustment costs—those who have economically viable low-carbon substitutes available to them or lower exposure to stranded assets—face weaker incentives to resist. Firms with high adjustment costs face stronger ones. Some low-adjustment-cost firms even support stricter regulation when tightening raises rivals' costs by more than their own (Kennard, 2020; Vormedal and Meckling, 2023).

Resistance operates through two sources of dirty-sector influence on policy that the business-power literature distinguishes (Culpepper, 2025). On the one hand, the dirty sector has structural power, which is rooted in firms' importance to the economy, as manifested by their Firms influence policy without direct action because policymakers anticipate adverse economic consequences from undermining business confidence (Lindblom, 1977; Przeworski and Wallerstein, 1988). The threat of relocation by multinationals is a particularly important channel, also in the climate context.[13] Instrumental power, on the other hand, refers to lobbying activities, which can be distinguished by the locus of action. *Inside lobbying* operates through direct contact with policymakers and bureaucrats: information, technical expertise, and framing of the policy problem in ways that structure elite discourse without becoming politically visible —what Culpepper (2011) calls "quiet politics".

is rife in climate policy (see e.g. Stokes (2020) on divisions among utilities and Fischer (2025) on the European car industry,). We do not deal with sectoral specificities given space constraints. But where intra-sectoral heterogeneity exists our framework can be applied at the sectoral level.

[11]Hedging refers to firms accepting climate regulation while protecting their core business either by shaping policy design to minimise compliance costs (Meckling, 2015) or by diversifying into low-carbon activities (Green et al., 2022).

[12]Kupzok and Nahm (2024) use the term "decarbonisable" to describe industries where adjustment is technologically and economically feasible. This is a subset of our category of low-adjustment dirty firms.

[13]Foreign-owned plants in the EU ETS receive systematically more lenient carbon regulation than the domestic plants of the same multinational, reflecting the greater credibility of relocation threats from operations with weaker country-specific commitments (Bayer, 2023).

It typically determines the technical design of instruments, such as the allocation mechanism for free allocations in the EU Emissions Trading System (Genovese and Tvinnereim, 2019; Sato et al., 2022; Melnick, 2024). *Outside lobbying*, by contrast, mobilises affected segments of the electorate, notably workers, in ways that make particular policy choices more or less electorally costly.[14] It is also the mechanism through which producers can affect the electoral signal, converting latent salience into noisy activation by promoting cost, jobs, or regional-loss frames (Stokes, 2020; Patashnik, 2023).

Adjustment, by contrast, integrates market and non-market strategies in the sense developed by Baron (1995, 2003). On the market side, firms invest in clean alternatives—capital expenditure on renewables, electric-vehicle platforms, low-carbon industrial processes—reposition business models away from carbon-intensive activities, and try to capture first-mover advantages. These include learning-curve-driven cost reductions (Breetz et al., 2018), early access to scarce inputs, the cultivation of relationships with reliable and high-quality suppliers, and the compliance know-how that comes from being first under a new regulatory regime (Dugoua and Moscona, 2025). On the non-market side, adjusters lobby to extract support or compensation for adjustment from the government and, as Kennard (2020) shows, they may indeed lobby *for* stricter regulation when doing so raises rivals' costs by more than their own. The market and non-market sides reinforce each other. Market repositioning gives firms standing to lobby for tighter standards, and successful non-market engagement raises competitors' costs in ways that increase the returns to market repositioning (Kim, 2022). There is thus a distinct political logic to adjustment, whereby firms with a competitive interest in tighter regulation become political allies of climate action against firms with a competitive interest in resistance.

Whether a firm resists or adjusts depends on several factors—of which we highlight four especially prominent ones. The first is technology. As clean alternatives mature and cheapen, the calculus tips towards adjustment (Breetz et al., 2018). The second is corporate governance. Political positions can diffuse across firms through shared board memberships, with firms whose directors also sit on the boards of

[14] Wolton (2021) formalises how inside and outside lobbying interact. For groups defending the status quo, the credible threat of outside lobbying induces policymakers to compromise on policy content, with inside lobbying serving as a means of communicating mobilisation capacity or outside lobbying power to the policymaker. For change-seeking groups, outside lobbying serves as a signal about the electoral appeal of reform or change to the policymaker.

companies that have already taken climate action becoming themselves more likely to support climate regulation (Lerner and Osgood, 2023). Boards dominated by investors with short-time horizons face the opposite pressure (Finnegan and Meckling, 2026). The third is reputational. Where consumer or investor preferences are pro climate, there are returns to taking some climate action—including corporate climate targets for firms (Dietz and Hastreiter, 2026). These include higher demand from green-conscious consumers, cheaper capital from ESG-oriented investors, and improved recruitment from employees who prefer sustainable employers (Meckling, 2015; Abito et al., 2019). The fourth factor concerns the coordination infrastructure available to firms. Business associations are the most widely studied component of this infrastructure (Mišić, 2025; Vesa et al., 2025; Jensen, 2025). Beyond pooling resources policing defection, and thus lowering the cost of coordinated resistance, they moderate the reputational factor. That is, they allowing lagging firms to "hide in the crowd" as outsiders cannot reliably attribute association lobbying to specific members (Toenshoff, 2025). Supply-chain integration enables coordination through a different channel. Upstream suppliers, downstream buyers, and firms locked into long-term contracts share commercial stakes in continued carbon-intensive production, and these stakes generate lobbying coalitions that operate across sectoral boundaries (Cory et al., 2021; Meckling, 2011).

All four factors are conditioned by policy credibility—whether firms believe policies and structural conditions will persist for long enough for their investments to pay off.[15] Where credibility is low, firms delay clean R&D, avoid sunk costs, and hedge (Bloom et al., 2007; Noailly et al., 2022; Basaglia et al., 2025). Accordingly, credibility is partly endogenous to the framework's electoral and producer variables. Policymakers' incentives to sustain or reverse prior commitments depend on the electoral signal and the producer configuration they confront, making the analysis of policy feedbacks central, as set out in Section2.3.

Overall, these factors capture the firm-level calculus, rather than a sectoral outcome. Two firms in the same industry can occupy at opposite ends of the resist–adjust continuum, and the same firm can move along it as structural conditions

[15]Gazmararian and Tingley (2026) distinguish two dimensions of credibility. *Strategic* credibility concerns firms' expectations about the political failures of credibility, while *structural* credibility concerns whether the underlying material conditions—technology maturity, supply-chain capacity, comparative advantage—are congenial. Both must hold for adjustment. A strategically credible commitment fails if structural conditions are absent, and structurally favourable conditions fail if firms expect reversal before payoff.

change. The central political question for our framework is how individual firm choices aggregate.

2.2.3 Dirty-sector (dis)unity

Aggregating individual firm choices yields our explanatory variable of interest on the producer side, to wit the degree to which dirty producers are unified or fragmented. The dirty sector is unified when most firms resist; it is fragmented when enough firms have defected toward adjustment that the sector no longer speaks with one voice and clean firms can join forces with adjusting dirty firms (Drahos, 2025). The climate politics literature documents that fragmentation can take different forms,[16] but we do not develop these distinctions further here.

What matters for our purposes is that sectoral (dis)unity does not simply reduce to the sum of firm-level preferences. Firms' positions depend on those of other firms. Firms with relatively high willingness to adjust may nevertheless oppose climate policy to maintain alliances. Conversely, firms with anti-climate preferences may break ranks, either to realise first-mover advantages or to avoid being left behind when they expect others to do the same (Drahos, 2025). Unity can be thus sustained even when underlying preferences are heterogeneous—and coordination can fail even when preferences are relatively homogenous. Finally, the shift from unity to fragmentation is often non-linear (Dugoua and Dumas, 2024), with a critical mass of defectors flipping the configuration rather than gradually eroding it.

The configuration matters because it shapes the policymaker's strategy space. It is itself partly endogenous. Past policy choices and the policymaker's current strategy reshape which firms mobilise and on which side. We turn to those strategies next.

2.3 The policymaker's strategy space

The electoral and producer sides specify the pressures the policymaker faces, but they do not yet specify what the policymaker does in response. To work out this best response, we first have to define policymakers' strategy space —the levers they can

[16] Active support generates new defenders of stringency, with clean firms or low-adjustment-cost incumbents lobbying for policy continuation. Acquiescence reduces opposition without creating new advocates; threatened losers accept compensation, exit bargains, or procedural guarantees and stop blocking. For hedging, see above.

use to navigate these political pressures. The variety of climate-policy instruments is vast: carbon taxes, emissions-trading schemes, performance standards, mandates, feed-in tariffs, public procurement, infrastructure investment, just-transition packages, R&D subsidies. The framework groups them into two ideal-typical strategies, accommodation and transformation, distinguished by the temporal horizon over which they operate and the political function they serve. Real instruments typically sit on a continuum between these poles rather than at either extreme.

The two strategies differ in what they aim to do and when their political payoff materialises. They are strategic complements more often than substitutes: pursuing one typically raises the political return to pursuing the other, since accommodation manages the present-term costs that transformation would otherwise generate, while transformation reshapes the future political environment in which accommodation will be needed. [17] The complementarity holds where fiscal capacity is sufficient to fund both. Where fiscal capacity is constrained, the two strategies compete for the same scarce resources and shift toward substitutes, forcing the policymaker to choose between revenue allocated to compensation and revenue allocated to industrial policy.[18]

Accommodation manages the distributional consequences of climate policy in the current electoral term. It compensates losers, hides or diffuses costs, and phases implementation to reduce present political backlash without reshaping the producer configuration. Its forms include revenue recycling for carbon pricing, targeted compensation for affected workers and regions, and cost socialisation through general taxation or utility rates (Bolet et al., 2024; Edenhofer and Genovese, 2024; Finnegan, 2022b). The political payoff is immediate, but the producer configuration is unchanged at the end of the electoral term.

Transformation, by contrast, aims to reshape the producer configuration beyond the current electoral term, working through the policy feedback mechanisms discussed above. Forms include negotiated phase-outs and transition packages that fragment the dirty sector by buying acquiescence (Meckling and Nahm, 2022; Kelsey, 2018; Mildenberger, 2020), industrial policy and subsidies that build clean-

[17] Bolet et al. (2024) show that Spain's coal-region just-transition agreement reduced electoral backlash to climate policy in affected districts; survey-experimental work corroborates the packaging logic (Fesenfeld, 2022; Wicki et al., 2020).

[18] The same instrument can serve both ends. Compensation for a coal region simultaneously accommodates current workers and transforms the producer configuration through managed exit (Bulow et al., 1985).

sector firms and electoral beneficiaries (Meckling et al., 2015; Stokes, 2020; Aklin and Urpelainen, 2018), and mandates, performance standards, or phase-out timelines that force firm-level adjustment through credible commitment (Meckling and Nahm, 2019).

Because transformation's payoff lies beyond the current term, it depends constitutively on credibility (Brunner et al., 2012; Nemet et al., 2017; Gazmararian and Tingley, 2026). Firms invest in adjustment, and constituencies form around new policies, only if actors expect commitments to persist (Bloom et al., 2007; Noailly et al., 2022). Ambitious instruments without credible commitment produce little behavioural response, and therefore little change in the producer configuration that conditions future policy choice. Credibility-building is thus not a separate strategic axis but an integral component of transformation itself. It operates through institutional commitment devices such as climate framework laws, independent advisory bodies, and statutory targets (Helm et al., 2003; Brunner et al., 2012; Finnegan, 2022b; Gazmararian and Tingley, 2026), and through the deliberate cultivation of constituencies whose interests would be harmed by reversal (Pierson, 2000; Patashnik, 2008; Meckling et al., 2015; Biber et al., 2018; Stokes, 2020).[19] The credibility of transformation commitments that depend on sustained fiscal flow varies further with the fiscal capacity that backs them, a moderator developed in Section 3.2. Crucially, credibility is itself partly endogenous to the transformation strategy: each transformation instrument that successfully induces investment, builds constituencies, or fragments the dirty sector raises the political cost of subsequent reversal, which in turn raises the credibility of further transformation commitments. This recursive structure is what makes transformation cumulative, and it is the bridge to the dynamic theory developed in Section 3.2.

Accommodation pays off in the current term, but transformation does not, which raises the question of why a myopic policymaker would ever pursue it. At least three mechanisms reconcile transformation with myopia. First, transformation instruments produce present-term benefits visible to voters — regional jobs, identifiable beneficiaries, infrastructure groundbreakings. Second, they realign electoral resources by creating clean-leaning constituencies whose support flows to enacting

[19]Climate institutions can serve as commitment devices, with institutional routes varying by electoral logic (Lockwood, 2021); see also Edenhofer and Flachsland (2025); Zwar et al. (2024). Policy can fail to persist through outright reversal or through drift, where formal rules remain but their substantive effect erodes as conditions change without updating (Hacker, 2004; Krehbiel, 1998). Drift is the failure mode characteristic of stagnation, developed in Section 4.

coalitions. Third, they secure political position in anticipation of future competitive dynamics: claim-staking commitments produce reputational returns in the present even when costs fall on successors (Jacobs, 2011a), and they position the policymaker against anticipated near-term pressure from green parties or protest cycles (Meguid, 2005; Spoon et al., 2014).[20] Transformation is therefore consistent with myopic policymaking when the political returns in the present are sufficient.

3 Configurations and trajectories

The previous section developed the two independent variables on which the framework's analysis hinges: the type of electoral demand—latent, noisy, or clear—and the configuration of the dirty sector—unified or fragmented. We now interact them, examining what their combination implies for the durability of climate-policy stringency. The interaction yields six political configurations,[21] and a set of trajectories along which countries move as configurations shift over time.

In what follows, country-level claims should be construed as shorthand for the policy episodes during which durable stringency is being built, contested, or reversed. The relevant unit of analysis is the policy episode itself, which sometimes operates at the country–sector level. This includes utility regulation, building codes, transport standards. However, on a country level, national instruments such as carbon prices, framework laws, and net-zero targets can bind across sectors.

We discuss these configurations in two steps. Section 3.1 sets out the six configurations, characterising in each case the strategic interactions between voters, policymaker, and producer. In Section 3.2, we turn to the dynamics: the sources of movement between configurations, the structure of plausible transitions, and the two recurrent trajectories towards durable stringency, namely cumulative transformation and contested accumulation.

[20] The dynamic political competition literature shows that parties accept present costs to shape the future political environment they will compete in (Izzo, 2023).

[21] We use "configuration" rather than "equilibrium", given that we do not solve a formally specified game.

3.1 The configuration space

The six configurations implied by our analytical primitives are summarised in Table 1.

Table 1: The six configurations of climate politics

	Producers unified	**Producers fragmented**
Latent demand	Quiet dirty dominance	Quiet producer transformation
Noisy activation	Noisy dirty dominance	Noisy contest
Clear activation	Weak electoral dominance	Strong electoral dominance

Rows distinguish the type of electoral demand the policymaker faces: latent (no climate signal), noisy, or clear. Columns distinguish whether the dirty sector acts as a coordinated bloc or is divided. The clean sector is not listed in the table, serving—as explained in Section XXXX—a counterweight to the dirty sector.

Given the type of electoral demand and the producer configuration, we now examine what strategy an office-seeking policymaker is likely to choose and what implications the latter has for the durability of climate stringency.

Quiet dirty dominance

This configuration combines latent demand with a unified dirty sector. Climate is not salient for most voters, and dirty producers are united in opposition to stringent climate policy. The policymaker therefore faces weak electoral incentives for action and strong producer pressure against it, implying low overall climate policy stringency.[22]

On the electoral side, latent public opinion means that avoided climate damages and other benefits of climate action are electorally close to irrelevant. So, voters do not reward policymakers for climate action. Nevertheless, they dislike costs, meaning they are ready to punish policymakers for pursuing policies that are costly for them, especially when these costs are clearly "traceable" (Arnold, 1990) to policy choices. Inaction has thus strong electoral appeal for office-seeking policymakers. The producer configuration reinforces this tendency. A unified dirty sector can

[22]Transformation is used only in ways and at margins that the dirty sector tolerates, possibly including (sub-national) climate policy when it is too insignificant to threaten dirty incumbents.

rely on structural power—whereby policymakers anticipate that stringent climate policy may affect investment, employment, energy prices, or relocation decisions—or (inside) lobbying to extract concessions or rents by shaping the design of policies, securing exemptions, or keeping threatening measures off the agenda (Lindblom, 1977; Culpepper, 2011).[23]

While accommodation usually means preserving the status quo—and doing so engenders no additional costs—cost-related considerations are still relevant in this configuration. Indeed, costs can arise in two circumstances. The first is when accommodating dirty producers via, for instance, subsidising industrial electricity consumption entails costs for voters. The second is when the policymaker adopts climate measures for reasons other than satisfying public opinion writ large, including geopolitical objectives or placating core constituencies within the party. In that case, accommodation typically takes the form of exemptions or compensation for affected firms. In both circumstances, the policymaker seeks to hide, delay, or diffuse the costs of accommodation for voters through opaque policy instruments, like surcharges embedded energy bills or similarly technical rules whose distributional effects are difficult to observe.[24]

Quiet producer transformation

This configuration combines latent demand with a fragmented dirty sector. Climate remains low-salience for most voters, so the policymaker receives little electoral reward for climate action and remains reluctant to impose costs on voters, especially when those costs are visible and traceable. Compared with quiet dirty dominance, however, dirty producers no longer present a united front. Some continue to resist, others hedge or acquiesce, and some have begun to adjust. Clean producers can therefore find allies among incumbents that were previously part of the dirty coalition. Fragmentation opens space for higher stringency, although durable stringent policy remains difficult.

[23]Outside lobbying is rarely used while climate remains latent. Its relevance lies in the threat of politicisation. The prospect that dirty producers could mobilise workers, regions, or consumers against policy change strengthens their ability to extract concessions through inside lobbying (Wolton, 2021).

[24]Clean producers or climate NGOs can discipline this accommodation only under certain conditions. They must raise climate salience, frame the issue around benefits rather than costs, and draw on latent opinion sufficiently pro-climate to generate a clear electoral signal once activated. If successful, the polity moves into an activated configuration.

Fragmentation should not be confused with strong producer demand for climate action. Active pressure arises only where adjusting incumbents and clean producers organise around policies that expand their markets or protect their transition investments. Where fragmentation takes the weaker form of hedging or acquiescence, it mainly softens opposition. Other motives for action then matter more, including policy-seeking objectives related to the need to satisfy core voters within the governing party.

The policymaker can therefore pursue limited transformation through instruments that reshape producer interests while keeping voter costs low, diffuse, or hard to trace. These include deployment subsidies, tax credits, public procurement, infrastructure investment, de-risking contracts, negotiated exits, and standards with long phase-ins. Such instruments often appear as energy or industrial policy rather than climate policy. If they induce investment, firms acquire clean assets, develop new capabilities, and become more dependent on policy continuation. These feedbacks can raise the cost of later reversal and increase credibility over time. If they fail to materialise, firms hedge, delay investment, or return to defensive lobbying. The configuration therefore produces modest stringency in the short run, but can contribute to durable stringency by gradually reshaping the producer configuration future policymakers confront.

Noisy dirty dominance

This configuration combines noisy activation with a unified dirty sector. Salience is high, having risen for any of the reasons set out in Section **??**, but the electoral signal lacks a clear direction. As established there, the noisy signal can arise from dispersed or polarised underlying preferences, from elite disagreement on framing, or from both. What follows here does not depend on which route applies. The point is that the policymaker faces high voter attention without a clear mandate.

The calculus this generates is genuinely different from the configurations under latent demand. Inaction is now politically costly, because voters who care about climate are paying attention and punish failure to act. Action is also politically costly, because voters who oppose action or who oppose visible costs are paying attention too. The cost-traceability constraint from latent demand still operates, but it bites harder, since more voters are watching and traceable losses now generate more electoral risk than they did when climate was latent. The policymaker is

caught between two sources of punishment, with no obvious way out through either action or inaction alone.

The unified dirty sector shapes how the policymaker deals with this electoral bind. Under quiet dirty dominance, the dirty sector held outside lobbying in reserve because the status quo was not under threat. Here it is. Outside lobbying becomes operationally important, with the dirty sector mobilising through media, regional and labour constituencies, and broader cost-of-living narratives. The substantive content of this lobbying is the framing battle. The primitives noted that elite disagreement on framing is one reason for a noisy signal, and the dirty sector here is one of the principal elites doing the framing. Where it succeeds, climate gets framed primarily in terms of costs — energy prices, household burden, regional employment risk — rather than in terms of climate action. The noisy signal's direction tilts toward inaction or against ambitious action, even where underlying voter preferences include substantial pro-climate sentiment.

Two factors can produce this configuration, with each illustrating how endogenous and exogenous activation interact with the dirty sector's framing capacity. In the first, the dirty sector itself activates the issue pre-emptively, raising salience while controlling the framing. The aim is to lock in cost-side framing before clean-sector or environmental-movement activation can establish climate as an issue on its own terms. In the second, an exogenous shock such as an energy price spike or a fossil-fuel supply disruption raises salience involuntarily, and the dirty sector captures the framing battle that follows. Either factor produces the same configuration, with the same constrained calculus.

The policymaker's response is correspondingly constrained. Action when it occurs is heavily disguised, framed in non-climate terms such as competitiveness, jobs, or energy security, and paired with cost-diffusion arrangements that hide the policy's connection to the cost-side narrative the dirty sector has established. Symbolic action is common — announcements and pledges that do not translate into substantive change. Reversal of prior policy is also common where the cost-side framing makes it politically cheap. Stringency under this configuration is low, with whatever substantive action occurs concentrated in instruments that escape the framing battle.

The producer side remains nominally unified, but the configuration is more unstable than quiet dirty dominance. High salience makes the dirty sector's coordination more demanding. Some firms within the sector — those facing technology cost

declines or with adjustment options — have material reasons to defect that grow stronger the longer the configuration persists. The framing battle keeps the sector unified longer than market incentives alone would, but the drift toward fragmentation is real. Over time, the configuration tends toward noisy contest, which is taken up next.

Noisy contest

Climate is salient and contested, but producers are fragmented. Dirty producers no longer fully control the producer side; clean firms, decarbonisable incumbents, and acquiescent former opponents partially counterbalance them. Neither public demand alone nor unified producer power determines outcomes, and the cell's defining feature is this rough parity between the two sides.

The framework's primitives support fewer determinate claims about policy output here than in the corner cells. With electoral and producer pressures cutting across each other, what gets adopted depends on which voters are pivotal and which producers organise most effectively. What the primitives suggest is a comparative claim about reversibility: producer-embedded commitments tend to be more costly to reverse than voter-facing instruments, because they create organised beneficiaries who can resist dispossession (Pierson, 1993).

The distributional pattern features both visible costs and visible benefits, with neither side rendering its preferred frame dominant. The diffuse beneficiaries of climate stabilisation acquire indirect representation only through producer-embedded coalitions.

Weak electoral dominance

Climate is clearly activated, but dirty producers remain unified. The office-seeker can identify a vote-maximising pro-climate direction, but unified producer resistance still shapes instruments, pace, exemptions, compensation, and implementation. The configuration is electorally favourable but politically incomplete: voters want action, producers shape what action looks like.

The result is high headline ambition combined with extensive producer accommodation. Targets and framework laws can be adopted at the level the electoral signal encourages, but the instruments delivering them are negotiated to preserve incumbent interests. Implementation gaps emerge where producer leverage is

strongest — typically in heavy industry, agriculture, and fossil extraction, where adjustment costs are high and coordination capacity is preserved.

The distributional pattern is one of visible ambition and concentrated producer cushioning. Household costs are diffused or compensated; producer benefits in the form of exemptions, free allowances, or transition support are technically obscured in design.

Strong electoral dominance

Climate is clearly activated and dirty producers are fragmented. Both pressures on the office-seeker point in the same direction: the electoral incentive favours action, and the dirty producer side has lost the cohesion needed to obstruct it. Where adjusters lobby actively for stricter regulation that raises rivals' costs, fragments of the former dirty coalition become positive supporters of stringency.

The result is high stringency, both adopted and implemented. The electoral signal makes adoption attractive, and the absence of unified producer resistance makes implementation feasible. Design choices reflect clean-coalition priorities as well as accommodation of incumbents, since clean and convertible producers have organised political weight on the producer side.

The distributional pattern features visible benefits to organised clean constituencies and accepted costs to households and dispersed incumbents. The most politically invisible interests are residual fossil constituencies whose decline is the configuration's implicit settlement.

Determinacy varies systematically across the configuration space. In quiet dirty dominance and strong electoral dominance, only one pressure is operative — producer in the former, electoral in the latter — and faces no opposing pressure capable of redirecting policy. The framework yields determinate predictions about policy output in these cells. In weak electoral dominance and noisy contest, both pressures are operative but cut across each other; outcomes turn on contingent dynamics that the configuration itself does not determine. The remaining two cells occupy intermediate positions, with one pressure operative and the other latent or divided.

3.2 Dynamics: from configurations to trajectories

The static cells of Section 3.1 describe what climate politics looks like in a sector–period while the underlying coalition holds. Coalitions break, however, and configurations change. This subsection sets out the dynamic theory: what moves a sector–period between configurations, which transitions are plausible, and through which trajectories enacted stringency becomes durable. The two trajectories of cumulative transformation and contested accumulation recur empirically and organise the cases in Section 4.

What moves a sector–period

Movement along the electoral dimension comes from three kinds of forces. Some are sudden and exogenous: natural disasters, energy-price shocks, court rulings, security crises, or protest waves can push climate or adjacent issues onto the political agenda. Others are gradual and exogenous: generational replacement, long-term value change, lived experience with climate impacts, and the diffusion of green norms alter the distribution of preferences over time (Besley and Persson, 2023). Still others are endogenous to political conflict, as parties, movements, firms, and media actors politicise climate, suppress it, or activate it through particular frames. Activation does not simply increase salience; it determines whether climate enters debate as a question of decarbonisation, household costs, energy security, or industrial opportunity.

Movement along the producer dimension also comes from three kinds of forces, though in different proportions. Sudden exogenous shifts are rarer than on the electoral side but possible: a major firm's strategic pivot, a coal company bankruptcy, the collapse of a trade association, or a foreign carbon-border adjustment can reorganise sectoral coalitions. Gradual exogenous change is the more common driver, as technology-cost declines, fossil-fuel prices, capital-market conditions, and foreign climate policies alter firms' incentives to resist or adjust. Endogenous policy effects make up the third source: subsidies, standards, infrastructure, phase-out dates, and compensation bargains can induce investment, create decarbonisable constituencies, split trade associations, or move losers from resistance to acquiescence.

Shocks differ not only in which side they act on but in the channel through which they operate. Salience-raising shocks, including natural disasters, energy-

price spikes, and security crises, push climate or adjacent issues onto the electoral agenda without specifying signal direction (Egan and Mullin, 2017); whether they activate consensual or noisy signals depends on which actors capture the framing battle that follows (Druckman, 2022). Coalition-disrupting shocks erode the conditions for unified producer coordination: a major firm's strategic exit, the collapse of a trade association, or a corporate scandal can fragment the dirty side by exposing firms to public scrutiny and undermining the reputational asymmetries on which quiet politics depends (Culpepper, 2011; Meckling, 2011; Vormedal and Meckling, 2023; Vesa et al., 2025). Credibility shocks, including court rulings, constitutional decisions, supranational commitments, and disruptive turnover at major emitters, alter what policymakers can credibly commit to or be forced to deliver, sometimes locking in stringency against subsequent reversal and sometimes unlocking it (Lockwood, 2021; Setzer and Higham, 2024; Colgan and Genovese, 2025). The same triggering event can operate through more than one channel: the Ukraine energy crisis raised salience around costs in some countries while functioning as a credibility shock in the EU, where political response steered private capital toward clean alternatives (Bayer et al., 2025).

Producer movement tends to be slower than electoral movement, but more durable once it occurs. The temporal asymmetry is the load-bearing feature of the dynamic theory: salience can rise and collapse within an electoral cycle, while investment, infrastructure, supply chains, firm adjustment, and constituency formation take time. Durable stringency depends on whether supportive feedback materialises before salience fades, costs are reframed, or opponents return.

Adjacency. Not every transition is equally plausible. Sector–periods most often move to adjacent cells, changing position along one dimension at a time, because the underlying primitives shift along their own dimensions before reorganising jointly. A sector in quiet dirty dominance can move to quiet producer transformation (producers fragment under latent demand) or to noisy dirty dominance (salience rises while producers remain unified) more easily than it can leap to strong electoral dominance. Weak electoral dominance can advance to strong electoral dominance if producers fragment, or decay into noisy dirty dominance if rising costs let unified producers reframe climate policy as cost imposition. Non-adjacent transitions require larger shocks: a technology breakthrough that simultaneously fragments producers and activates voters, or a crisis that reshapes both sides at once.

Two routes to durable stringency

Two trajectories recur in the empirical record. Both rely on the policy feedback mechanisms developed in Section 2.3: the policymaker's transformation strategies reshape the producer configuration and electoral landscape in which later policy is chosen (Stokes, 2020). The trajectories differ in the electoral conditions under which feedback must materialise, in their characteristic time horizons, and in their characteristic failure modes. Both are paths through the configuration space; both depend on the temporal asymmetry above, but they exploit it in opposite ways. Whether feedback materialises turns on two dimensions of credibility (Section **??**): *strategic* credibility, whether enacted policy survives long enough to generate feedback, and *structural* credibility, whether material conditions allow it to deliver the jobs, supply chains, and constituencies on which feedback depends (Gazmararian and Tingley, 2026).

Cumulative transformation. Policy feedback accumulates across repeated episodes under permissive electoral conditions, gradually weakening dirty-sector cohesion or creating electorally relevant beneficiaries. The trajectory begins under latent demand or clear activation, where the office-seeker faces no hostile-alternation risk that would force fast feedback. Under latent or low-salience demand, climate-relevant policy is justified through non-climate rationales — industrial strategy, energy security, local development, air pollution, innovation — and operates through the back door of quiet producer transformation. Under clear activation, the office-seeker can identify a vote-maximising pro-climate direction and adopt policy at the level the electoral signal encourages.The trajectory requires both dimensions of credibility to hold over a long horizon, but neither must materialise quickly.

What matters is not the initial level of salience but whether policy creates feedback effects that survive the episode in which the policy was adopted. Three feedback subtypes recur. *Producer feedback* operates when policy creates clean firms, supply chains, employment, and lobbying capacity that fragment dirty-sector cohesion and create political defenders of continuation. *Consumer and infrastructural feedback* operates when policy creates mass user constituencies and infrastructure that lower the cost of subsequent adoption. *Institutional reinforcement* operates when statutory targets, advisory bodies, or carbon budgets make drift visible and structure subsequent debates. The trajectory's characteristic failure mode is *stagnation*:

policy persists formally, ambition remains institutionalised, but feedback effects are too thin or sectorally isolated to reshape future politics. Stagnation is not the absence of policy but the failure of policy to change the future strategy space — a country may preserve statutory targets while implementation stalls in sectors where producer adjustment, consumer adoption, infrastructure, or electoral support remain underdeveloped (Stokes, 2020). The trajectory demands fast feedback on both dimensions: hostile electoral conditions raise the strategic-credibility bar, while compressed time horizons raise the structural-credibility bar.

Contested accumulation. Under noisy activation, public demand alone does not determine what the office-seeker should do: climate action rewards some voters and alienates others, and opponents can frame policy around costs, jobs, prices, or regional loss. Noisy activation therefore does not predict climate policy — it predicts underdetermination unless an electoral, coalitional, institutional, or producer-side anchor pins down policymaker incentives. The anchor can take several forms: a core pro-climate constituency, a coalition partner, a pivotal region, a court or constitutional obligation, an EU or international commitment, party ideology, clean or decarbonisable firms, local governments, or affected workers and communities that value a bargain. Anchoring is necessary but not sufficient: because noisy activation carries a high risk of alternation, supportive feedback must materialise faster than opponents can return.

Two feedback subtypes recur. *Opposition-dampening feedback* moves unified resistance toward acquiescence: losers do not become climate supporters, but compensation, procedural guarantees, regional investment, or exit bargains reduce the value of continued obstruction. *Constituency-building feedback* creates new electorally relevant defenders of continuation through investment, manufacturing capacity, supply chains, local employment, and place-based incentives. The trajectory's characteristic failure modes are *reversal* and *dilution*: opponents return before supportive feedback becomes politically protective, or visible costs mature before benefits and adjustment pathways become credible. The asymmetric reversibility of producer-embedded versus voter-facing commitments (Pierson, 1993) is what gives contested accumulation analytical bite — producer-embedded commitments resist reversal even when voter-facing instruments are repealed.

Moderators and the net-zero endgame

Alongside the shocks discussed above, slower-moving structural features moderate the rate and probability of movement between configurations rather than defining additional trajectories. The most consequential of these are institutional. Electoral systems, corporatist intermediation, and federal structure each shape how the framework's primary variables produce durable stringency or fail to do so. None of them, however, generates a directional prediction on its own. Each is two-edged, amplifying and dampening pressures in ways that depend on which configuration obtains. This is why the framework treats institutions as moderators rather than primary explanatory variables.

Electoral institutions are theorised to moderate the bindingness of the cost-traceability constraint: where vote-seat elasticities are low, governments can absorb concentrated electoral punishment for traceable cost imposition, relaxing the constraint that otherwise pushes policymakers toward accommodation under noisy activation(Finnegan, 2022a; Lipscy, 2018). The same proportional rules, however, can also create coalition-level veto points through which pivotal partners can block or dilute policy, and they lower the threshold for niche-party entry: green-party success can prompt mainstream parties to de-emphasise climate rather than match its ambition, since established parties have weaker incentives to compete on an issue an entrant owns (Abou-Chadi, 2016; Abou-Chadi and Kamphorst, 2023). Majoritarian systems have produced some of the most durable framework legislation when alternative anchors compensate for weaker insulation (Lockwood, 2021; Wiedemann, 2024). On the producer side, corporatist interest intermediation provides venues for negotiating compensation packages, phase-out timelines, and procedural guarantees that produce opposition-dampening feedback under contested accumulation (Meckling and Nahm, 2022; Finnegan, 2022b). The same venues, however, give organised carbon-intensive interests privileged access to policymakers, double-representing dirty-sector preferences through both business and labour channels and entrenching resistance where corporatist confederations remain tied to fossil sectors (Mildenberger, 2020).Federalism opens subnational venues where cumulative transformation can begin under nationally hostile configurations and disperses electoral stakes in clean-energy investment across legislative coalitions in ways that anchor support for continuation (Stokes, 2020; Aklin and Urpelainen, 2018). The same federal structure, however, provides regionally concentrated dirty sectors

with institutional venues to block national stringency, especially where carbon-intensive production is geographically concentrated in politically pivotal jurisdictions (Stokes, 2020; Mildenberger, 2020).

Fiscal capacity is a more directionally clear moderator: it shapes which commitments are credible and which trajectories are reachable (Meckling and Benkler, 2024): cumulative transformation depends on sustained subsidies and infrastructural investment that fiscally constrained states can run only weakly, slowing feedback accumulation, while contested accumulation depends on opposition-dampening payments or constituency-building investment that fiscally constrained states cannot deploy at scale. Fiscal constraint therefore narrows the strategy space asymmetrically, biting harder on transformation than on accommodation, since accommodation can be diffused through general taxation or utility rates while transformation typically requires discretionary outlays (Juhász and Lane, 2024).

Technological change is the most cumulative of these moderators: falling clean-technology costs simultaneously lower firm-level adjustment costs, raise the structural credibility of clean-investment commitments, and ease consumer adoption, compounding across episodes in a way that other moderators do not (Nemet, 2006; Nemet et al., 2017). Cost declines therefore favour producer fragmentation through multiple channels at once, accelerating cumulative transformation where it has begun and lowering the political cost of opening noisy cases (Meckling et al., 2015; Meckling and Nahm, 2018).

The framework predicts that movement through the configuration space becomes more demanding as climate policy extends into harder sectors. Heating, buildings, heavy industry, agriculture, land use, and fossil extraction expose policymakers to more visible household costs, more concentrated regional losers, and stronger opportunities for opponents to frame policy as unfair or infeasible. Sector–periods in these domains are more likely to occupy noisy cells. Cumulative transformation becomes harder to engineer because non-climate justifications are less available; contested accumulation becomes the trajectory on which durable stringency increasingly depends. The cases in Section 4 illustrate both trajectories under varying conditions, and Section 5.1 draws out the implications for the future politics of net zero.

3.3 Summary

The framework identifies two routes to durable climate-policy stringency. *Cumulative transformation* accumulates feedback across repeated episodes under permissive electoral conditions, gradually weakening dirty-sector cohesion or creating electorally relevant beneficiaries; its characteristic failure is stagnation. *Contested accumulation* relies on an electoral, coalitional, institutional, or producer-side anchor under noisy activation, with supportive feedback materialising faster than opponents can return; its characteristic failures are reversal and dilution.

Table 2: Trajectories, feedback requirements, and characteristic failures

Trajectory	Electoral condition	Feedback requirement	Characteristic failure
Cumulative transformation	Latent or clear incentives are sufficient for policy adoption	Feedback accumulates across repeated episodes, weakening dirty-sector cohesion or creating electorally relevant beneficiaries	Stagnation: policy persists formally but does not reshape future politics
Contested accumulation	Noisy activation plus an electoral, coalitional, institutional, or producer-side anchor	Fast policy feedback dampens opposition or builds constituencies before opponents return	Reversal or negative feedback: costs and counter-framing mature before supportive feedback

Section 4 applies the framework to selected country–sector–policy episodes, organised by trajectory and feedback subtype. Cumulative transformation is illustrated through producer feedback (Denmark, German renewables), consumer and infrastructural feedback (Norway), institutional reinforcement (UK power), and stagnation (UK harder sectors). Contested accumulation is illustrated through opposition-dampening feedback (Germany coal), constituency-building feedback (the US Inflation Reduction Act), reversal (Australia), and dilution (Germany heating). Section 5.1 draws out the implications for public opinion, business power, policy design, sequencing, and the future politics of net zero.

4 Empirical illustrations of trajectories

This section applies the framework to selected country–sector–policy episodes. The objective is analytical rather than exhaustive. Climate-policy trajectories are complex, and a systematic qualitative comparative analysis would require explicit coding of electoral demand, producer configuration, framing, institutions, policy instruments, and feedback effects across many sectors, countries, and years. That is beyond the scope of this article. The cases below therefore provide a first-order approximation. They are intended to illustrate the leverage of the framework: how the same primitives help distinguish trajectories to durable stringency, forms of feedback, and characteristic failures.

For the empirical illustrations, durable stringency is conceived as climate policies that impose binding costs or mandates at adoption and retain those binding parameters across at least one government turnover involving credible repeal pressure. The characteristic failure modes follow: reversal (formal repeal), dilution (formal survival with actively weakened parameters), drift (formal survival but erosion of substantive effect as conditions change without policy updating), and stagnation (survival without sectoral change). The durability window is case-specific, anchored to adoption and the relevant turnover episodes. Sectoral emissions are downstream and jointly determined by policy, technology, and prices; the cases below treat them as suggestive rather than defining.

The case-selection strategy follows from this objective. In the language of case-study methodology, the cases are selected for theoretical coverage and mechanism illustration rather than representativeness (Gerring, 2017; Humphreys and Jacobs, 2015). Coverage is defined across two dimensions: the framework's two trajectories and the feedback subtypes within each. The cases also span sectors (electricity, transport, heating, heavy industry, fossil extraction) and institutional contexts (majoritarian and proportional systems, federal and unitary states, corporatist and pluralist intermediation), allowing the moderators developed in Section 3.2 to vary across cases. Table 3 summarises the resulting selection.

4.1 Cumulative transformation

Quiet, cumulative transformation occurs when policy feedback accumulates across repeated episodes, gradually changing the producer, consumer, institutional, or in-

Table 3: Case selection and framework variables

Case	Electoral demand	Producer configuration	Trajectory	Feedback subtype	Outcome
Denmark wind (1979–)	Latent	Fragmenting	Cumulative	Producer (export)	Durable
Germany EEG (2000–)	Latent to noisy contest	Unified to fragmenting	Cumulative	Producer (mass deployment)	Durable
Norway EVs (1990s–)	Latent	No domestic ICE coalition	Cumulative	Consumer/ infrastructural	Durable
UK power (2008–)	Clear activation	Fragmenting	Cumulative	Institutional plus sectoral	Durable
UK harder sectors (2008–)	Clear to noisy	Unified	Cumulative	None mature	Stagnation
Germany coal (2018–2020)	Noisy activation	Unified at outset	Contested	Opposition-dampening	Durable
US IRA (2022–2025)	Noisy contest	Fragmenting	Contested	Constituency-building (uneven)	Reversal
					Australia CPM (2012–2014)
Noisy dirty dominance	Unified	Contested	Insufficient	Reversal	
France carbon tax (2014–2018)	Noisy dirty dominance	Household cost dominant	Contested	Insufficient	Reversal
Germany heating (2023–)	Noisy activation	Fragmented but thin	Contested	Insufficient	Dilution

Note: Key moderators vary across cases. Cumulative-transformation cases turn on small-economy export orientation (Denmark), federal corporatist intermediation (Germany EEG), fiscal capacity and sectoral separation between end-use and extraction (Norway), and majoritarian framework legislation (UK). Contested-accumulation cases turn on corporatist compensation venues (Germany coal), federal place-based industrial policy (US IRA), majoritarian alternation against mining-dependent economies (Australia), fuel-price visibility under diesel dependence (France), and coalition politics interacting with supply-chain readiness (Germany heating).

frastructural conditions under which later policy is chosen. It can begin under latent demand, when climate-relevant policies are justified through industrial strategy, energy security, innovation, local development, or consumer benefits. It can also be reinforced under clear activation, when policymakers can identify a vote-maximising pro-climate direction. The common feature is that feedback does not have to materialise immediately to prevent reversal. Because the electoral signal is latent or broadly supportive, policy has more time to reshape the future strategy space. The trajectory's characteristic failure is stagnation: institutions and ambition persist, but sectoral feedback remains too thin to reshape future politics. The cases below illustrate the trajectory's three feedback subtypes (producer, consumer/infrastructural, and institutional) and the stagnation failure mode.

Producer feedback through export manufacturing.

Denmark wind (1979–). *Configuration at adoption:* latent demand, fragmenting producers. *Anchor:* industrial-policy and energy-security rationale after the 1973 oil shock. *Feedback mechanism:* export-oriented wind manufacturing creates clean-sector firms, supply chains, and employment with electoral and lobbying capacity. *Outcome:* durable.

Denmark's wind trajectory operates in a configuration close to quiet producer transformation. Electoral demand for climate policy was not continuously activated, but the producer side gradually changed. Early support was framed through energy security after the 1973 oil shock and industrial development for a small open economy. Wind policy created firms, supply chains, local employment, export interests, grid-integration capabilities, and increasingly organised clean-sector constituencies. These feedback effects altered the producer configuration before climate became a sustained electoral cleavage. Later activation therefore occurred in an electricity sector in which clean producers and local beneficiaries could counterbalance fossil incumbents. Producer feedback here ran through a manufacturing sector rather than a deployment subsidy; the export orientation gave Danish wind firms commercial stakes that did not depend on continued domestic subsidy at any single point in time.

Producer feedback through mass domestic deployment.

Germany EEG (2000–). *Configuration at adoption:* latent demand transitioning toward noisy contest, unified producers fragmenting. *Anchor:* industrial-policy and ecological-modernisation rationale with Red-Green coalition support. *Feedback mechanism:* cost-diffusing surcharges fund mass deployment, creating wind and solar firms, municipal actors, and local investors with stakes in continuation. *Outcome:* durable.

Germany's renewable-electricity trajectory exhibits a similar feedback logic through a different policy design. The *Erneuerbare-Energien-Gesetz* created concentrated beneficiaries through cost-diffusing support for renewable electricity: wind and solar firms, local investors, municipal actors, employment, grid requirements, and clean-sector lobbying capacity. The starting configuration was close to quiet dirty dominance in electricity, with fossil and utility incumbents still powerful. Over time, renewable support generated producer feedback that moved the sector toward quiet producer transformation and, later, noisy contest or stronger electoral dominance in electricity. Producer opposition did not disappear, but it became less unified.

This interpretation aligns with work on competing interest groups and policy–industry feedback. Cheon and Urpelainen (2013) show that renewable-supporter coalitions can increase renewable electricity generation, but that their effect depends on the strength of industrial opposition. Kelsey (2018) similarly argues that policy feedback is most likely where affected industries include winners or convertible actors able to alter their business models in response to early policy. The two episodes therefore illustrate the same feedback subtype operating through different mechanisms: an export industry in the Danish case, mass domestic deployment with cost-diffusion in the German. Both demonstrate the core mechanism of cumulative transformation: policy changes the producer configuration, and later activation occurs on more favourable terrain.

Consumer and infrastructural feedback.

Norway EVs (1990s–). *Configuration at adoption:* latent demand, no domestic ICE producer coalition, sectorally separated from fossil extraction. *Anchor:* early industrial-policy interest in EV manufacturing, later consumer-cost rationale. *Feedback mechanism:* tax advantages, road-use benefits, and charging infrastructure create mass adopter constituency with stakes in continuation. *Outcome:* durable.

Norway's electric-vehicle rollout illustrates cumulative transformation through consumer and infrastructural feedback. The relevant episode is domestic passenger transport, not Norwegian climate politics as a whole. Electoral demand was

permissive rather than continuously activated around a clear national climate mandate. Producer configuration was also unusual: Norway lacked a powerful domestic internal-combustion vehicle producer coalition, while EV policy did not directly threaten the petroleum-export coalition. This sectoral separation matters. Norway could advance domestic end-use electrification while fossil extraction remained politically shielded, a pattern consistent with Mildenberger's account of the double representation of carbon-intensive interests (Mildenberger, 2020).

The feedback mechanism in passenger transport operated through consumers and infrastructure. Long-running tax advantages, road-use benefits, charging infrastructure, and rising consumer adoption created a mass user constituency with material stakes in continuation. Adoption lowered social and informational barriers for subsequent adopters, while infrastructure reduced the practical cost of electrification. The episode therefore extends the framework beyond producer feedback. Public opinion matters here not only through voting, but through technology adoption and siting, which shape the political economy of the clean-energy transition directly (Gazmararian et al., 2025). Norway shows that end-use transformation can move far ahead of fossil-producer transformation when the targeted policy does not strand the dominant fossil coalition's assets. An obvious alternative reading is that fiscal capacity from the petroleum fund subsidised EV adoption directly, with consumer feedback as a downstream rather than causal mechanism. The framework's response is that fiscal capacity was necessary but not sufficient: other resource-rich economies have not generated comparable end-use transformation, and the sectoral-separation condition (no domestic ICE coalition with veto power) was what allowed sustained subsidy without producer counter-mobilisation. Fiscal capacity and sectoral separation operate jointly rather than as substitutes.

Institutional reinforcement and sectoral feedback.

UK power (2008–). *Configuration at adoption:* clear activation, fragmenting producers. *Anchor:* cross-party convergence around the Climate Change Act, supported by NGO mobilisation and elite consensus. *Feedback mechanism:* statutory targets and Climate Change Committee combined with electricity-market reform, contracts for difference, and offshore-wind deployment. *Outcome:* durable.

The UK power sector illustrates cumulative transformation through institutional reinforcement paired with material sectoral feedback. The electoral configuration around the 2008 Climate Change Act was close to clear activation. Major parties

converged around the need for climate action, and policymakers could identify a vote-maximising pro-climate direction. The Act created institutional feedback through statutory targets, carbon budgets, and the Climate Change Committee. These institutions made drift visible, structured subsequent policy debates, and increased the reputational cost of abandoning ambition.

Power-sector durability, however, depended on the interaction between institutional and material feedback. Electricity-market reform, contracts for difference, offshore-wind deployment, coal phase-out, and grid investment created producer and infrastructural effects that changed the future strategy space. Clean and convertible actors gained material stakes in continuation, while coal-fired generation became less central to the electricity coalition. The sector therefore moved toward strong electoral dominance: a broadly pro-climate electoral signal combined with a producer configuration increasingly supportive of, or at least less unified against, decarbonisation. An institutionalist reading would attribute durability to the Climate Change Act alone. The next case suggests otherwise: the same institutional framework has not prevented stagnation in heating, buildings, and industry, which indicates that institutions matter most where sectoral feedback also develops.

Climate institutions can stabilise expectations and force attention back onto the agenda, but their effects are strongest where sectoral feedback also develops. Recent institutional work makes a similar point: institutions facilitate transformation through mechanisms such as insulation and compensation, but their effects vary across demand-side and supply-side policies and across sectors (Finnegan et al., 2025).

Stagnation as failure mode. The UK's harder sectors illustrate stagnation, the characteristic failure of cumulative transformation. The country retains a strong climate framework, statutory targets, and advisory capacity. Electoral demand has often been broadly pro-climate at the level of ambition, and institutional feedback continues to make implementation gaps visible. Yet heating, buildings, parts of transport, and industry have not generated feedback comparable to the power sector. Producer adjustment is thinner, consumer-facing costs are more visible, supply chains and installation capacity matter more, and the benefits of policy are less immediate.

The configuration in these sectors is therefore closer to weak electoral dominance or noisy contest than to strong electoral dominance. Policymakers retain

incentives to preserve ambition, but the producer and consumer feedback needed for deep implementation remains underdeveloped. Stagnation is the resulting outcome: ambition and institutions persist, while enacted sectoral stringency lags. This distinction matters for the dependent variable. Durable targets are not equivalent to durable enacted stringency. Stagnation marks the point at which sequencing has delivered sectoral depth in one domain without generating the repeated activation and feedback needed to broaden durable stringency across the economy.

4.2 Contested accumulation

Contested accumulation occurs when policymakers act under a divided or uncertain electoral signal because some anchor gives them reason to do so, and supportive policy feedback materialises quickly enough to constrain later reversal. This trajectory begins from a less forgiving political condition than cumulative transformation. Climate is salient, but the signal is divided. Opponents can frame policy around costs, jobs, prices, state overreach, or regional loss. Government turnover can reopen the settlement. Under these conditions, policy must generate fast feedback. It must dampen opposition or build electorally relevant constituencies before opponents return or negative feedback dominates.

Opposition-dampening feedback. Germany's coal phase-out illustrates opposition-dampening feedback. The starting configuration was weak or noisy dirty dominance. Climate was electorally relevant, but coal interests remained embedded in regional economies, utilities, unions, and state governments. Producer configuration was therefore closer to unified resistance than to supportive fragmentation. Coal exit threatened jobs, local fiscal capacity, regional identity, and incumbent assets. Policymakers had an anchor in Germany's climate commitments and in a broader electoral and institutional environment favourable to coal exit, but durable stringency required reducing the resistance of potential veto players.

The Commission on Growth, Structural Change and Employment linked coal exit to compensation, regional investment, worker protections, and procedural legitimacy. These instruments generated opposition-dampening feedback. Utilities, mining regions, unions, and regional governments faced a bargain whose reversal would reopen distributive and procedural commitments. The feedback effect was acquiescence rather than green conversion: coal actors did not become advocates of

climate policy, but unified resistance weakened.

This pattern fits broader work on reform capacity and compensation. Lindvall argues that reform in power-sharing settings depends less on the number of veto players than on institutional mechanisms that allow actors to solve commitment problems around side payments (Lindvall, 2010). Finnegan similarly emphasises how concertation and compensation can support long-term climate-policy investments (Finnegan, 2022b). Germany's coal phase-out therefore illustrates how contested accumulation can produce durable but imperfect stringency by dampening opposition. Durability in this case does not imply maximal stringency or net-zero adequacy.

Constituency-building feedback. The US Inflation Reduction Act illustrates constituency-building feedback under noisy contest. The electoral signal was divided, and hostile alternation was foreseeable. Producer configuration was mixed: fossil interests remained powerful, but clean firms, utilities, manufacturers, and low-adjustment-cost industries had grown more politically relevant. Policymakers acted with multiple anchors: Democratic core voters, clean-energy interests, low-adjustment-cost firms, unions, state and local development actors, and place-based industrial-policy incentives. The policy's trajectory depends on whether these anchors are reinforced by fast feedback.

The IRA's feedback mechanism is fiscal and material. It seeks to create clean and low-adjustment-cost beneficiaries through investment, manufacturing capacity, supply chains, local employment, consumer subsidies, and energy-community incentives. In framework terms, it attempts to move noisy contest toward producer fragmentation by making rollback costly for firms, workers, regions, and local governments. This is consistent with the decarbonisation-bargain literature: carbon-intensive yet decarbonisable, low-adjustment-cost industries may accept climate goals when policy de-risks transition and creates opportunities for green competitiveness (Kupzok and Nahm, 2024).

The case also shows why contested accumulation is fragile. Producer feedback must become electorally anchored. Firms may invest, but voters may fail to attribute local investment, jobs, or lower costs to federal climate policy. Benefits may be submerged, fragmented across levels of government, or too delayed to counteract opposition. Credibility is central. Firms, workers, and communities must believe that policy will persist before reorganising around it. Recent work on clean-energy

credibility emphasises this problem: governments must make compensation and green-investment promises believable if they are to reduce opposition and create durable climate coalitions (Gazmararian and Tingley, 2026). Climate-policy uncertainty can also depress investment and R&D among climate-exposed firms, weakening the feedback on which contested accumulation depends (Gavriilidis et al., 2026). The IRA is therefore a live test of whether fast, electorally anchored feedback can mature under noisy contest.

Negative feedback and reversal. Australia's Carbon Pricing Mechanism illustrates failed contested accumulation through negative feedback and reversal. The starting configuration was noisy dirty dominance. Climate was salient but divided, while mining, coal, and carbon-intensive export interests remained cohesive. Producer configuration was therefore unfavourable: dirty-sector actors could mobilise around a cost and competitiveness frame. The Gillard government had an anchor for action in its parliamentary coalition with the Greens and independents, but that anchor did not generate sufficient protection against electoral counter-mobilisation.

The policy included household compensation, free allocations for emissions-intensive trade-exposed industries, and supporting institutions. Yet the central instrument was a visible carbon price. Negative electoral feedback emerged quickly because costs were visible and attributable, while compensatory benefits were less salient. Producer-side feedback required a longer horizon: firms needed to believe the policy would persist before investing in adjustment, and the period before repeal was too short for major producer fragmentation. Opponents could frame the policy around household costs, economic harm, and broken trust before supportive feedback became politically protective.

The result was reversal. Some institutions and clean-energy capacity survived, but the central instrument was repealed before it reshaped the future strategy space. The episode fits Stokes's warning that policy feedback is contingent and contested. Policies expected to create a new politics can instead generate negative feedback when opponents mobilise during implementation through parties, public campaigns, regulators, courts, and lobbying (Stokes, 2020). Australia shows that under noisy activation, visible costs can mature faster than producer or voter-side benefits.

Negative feedback and dilution. Germany's heating-law conflict shows negative feedback and dilution rather than full reversal. The relevant domain was household heating. Electoral demand was noisy: climate was salient, but the dominant political conflict centred on household costs, feasibility, and state intrusion. Producer configuration was also unfavourable for rapid feedback. Heat-pump manufacturers, installers, energy-service firms, and local infrastructure actors existed, but they had not generated feedback comparable to the renewable-electricity sector. Supply-side capacity was constrained, consumer uncertainty was high, and benefits were not yet visible enough to offset fears of upfront burdens.

The policy had an anchor in the governing coalition's climate commitments, but the anchor weakened once opponents successfully framed the measure around household burden and administrative overreach. The outcome was dilution. Supportive feedback from installers, heat-pump supply chains, lower operating costs, and consumer normalisation had not yet materialised at sufficient scale. Negative feedback developed first.

This episode maps directly onto the framework's claim about noisy activation and fast feedback. Household-facing policies are especially vulnerable because cost pass-through, visibility, and attributability are high. Public-opinion research shows why this matters. Support for climate policy depends strongly on perceived effectiveness, perceived distributional consequences, and perceived household effects (Dechezleprêtre et al., 2025). Pocketbook effects are particularly salient for costly household-facing instruments such as carbon pricing and heating mandates (Beiser-McGrath and Bernauer, 2024). Germany heating therefore illustrates a failure mode likely to become more common as net zero moves into buildings, transport behaviour, food, land use, and local infrastructure.

Cross-trajectory comparison. The cases illustrate the framework's core claim: durable stringency depends on how electoral demand, producer configuration, framing, anchors, and feedback interact over time. cumulative transformation succeeds when feedback can accumulate across episodes without immediate hostile reversal. Denmark and German renewables show producer feedback; Norway EVs shows consumer and infrastructural feedback; UK power shows institutional reinforcement paired with material sectoral feedback. Stagnation occurs when feedback remains too thin or sectorally isolated, as in UK harder sectors. Contested accumulation succeeds only when policy is anchored and feedback becomes politically

protective quickly. Germany coal shows opposition-dampening feedback; the IRA attempts constituency-building feedback. Australia and Germany heating show the failure logic: negative feedback, cost framing, reversal, or dilution can occur before supportive feedback protects enacted stringency.

Cross-sectional moderators shape how these trajectories unfold. Institutional arrangements affect compensation, coordination, insulation, and credibility. Sectoral characteristics shape cost visibility, technology maturity, substitutability, and the ease of consumer adoption. Growth models and industrial composition determine whether clean or low-adjustment-cost firms can become politically relevant. Federalism and regional concentration affect whether place-based benefits become electorally anchored. Fossil-export dependence can shield extraction even while end-use sectors decarbonise, as Norway illustrates. These moderators help explain why similar instruments generate different feedback across cases.

Temporal moderators are equally important. Technology-cost declines can make producer fragmentation more likely; energy-price shocks can either strengthen cost backlash or reframe fossil dependence as a security problem; value change and generational replacement can alter the background distribution of preferences; policy uncertainty can delay investment; and repeated activation can force expansion into sectors that quiet sequencing leaves untouched. Timing is therefore central. Under clear activation, feedback has more time to mature. Under noisy activation, feedback must be fast, anchored, and credible. The next section draws out the implications for the future politics of net zero.

5 Taking stock and looking ahead

The framework developed above has a simple implication. Durable climate-policy stringency depends on whether policymakers can convert electoral incentives into policy feedback that changes the conditions under which later policy is chosen. The source of those incentives varies. Under latent demand, policymakers mainly serve core constituencies and organised interests while avoiding visible and attributable costs. Under clear activation, policymakers can identify a vote-maximising pro-climate direction. Under noisy activation, climate policy requires an anchor that gives office-seeking policymakers reason to act despite a divided signal. Across these settings, policy becomes durable when feedback reshapes producer strategies,

voter experiences, institutional routines, or material constraints before activation fades, costs are reframed, or opponents return. This section draws out the lessons for climate policymaking and then applies them to the future politics of net zero.

5.1 Lessons about climate policymaking

The first lesson is that salience is insufficient. For public opinion to constrain climate policymaking, activation must produce a signal that policymakers can interpret as electorally valuable. This amends the quiet-politics intuition. Low salience favours organised producer power, since policy is negotiated through technical channels with limited electoral scrutiny. High salience can constrain producer power when it clarifies a pro-climate direction. Yet high salience can also help opponents when it turns climate policy into a dispute over household prices, jobs, reliability, regional loss, or state intrusion. The electoral constraint on business power depends on what activated voters are understood to want.

This point connects directly to recent work on public demand. Climate concern in the abstract does not translate automatically into pressure on policymakers. Responsiveness depends on whether climate enters the electoral calculus in a form that rewards action and punishes delay. Salience helps when it combines with a clear opinion signal; salience alone can generate polarisation, backlash, or inaction (Schaffer et al., 2022). This also explains why survey support for climate action often overstates the political force of public demand. Many voters support climate goals while remaining uncertain about specific instruments, distributive effects, and personal costs. What matters for policymakers is therefore not public concern as such, but the electoral meaning of that concern.

The second lesson is that business power resides both in cohesion and in the ability to turn salience into contestation. Dirty producers are powerful when they coordinate around delay, dilution, exemptions, compensation, or reversal. This is the organisational side of business power. Climate politics also has a discursive side. Unified dirty producers can convert climate salience into cost salience. They can shift the issue from climate risk to household burden, from industrial opportunity to job loss, from energy transition to reliability, and from long-term gains to immediate sacrifice. In this way, producer power can operate through public activation rather than only through quiet influence.

The limit to this power is fragmentation. Climate politics changes when busi-

ness no longer speaks with one voice. Clean firms, renewable producers, lower-adjustment-cost firms, decarbonisable incumbents, local beneficiaries, and consumers with stakes in clean technologies can give policymakers alternative producer and voter constituencies. Producer fragmentation can take different forms. Some firms actively support stringency because it expands markets for clean goods and services. Others hedge because they expect regulation and seek to shape design. Some losers accept compensation and move from resistance to acquiescence. These shifts matter because they reduce the ability of dirty producers to present climate policy as an attack on business or workers as a whole.

This lesson builds on work showing that firms choose between opposition, support, hedging, and non-participation depending on distributional effects and regulatory pressure (Meckling et al., 2015). It also fits evidence that renewable supporters can matter when opposition is weak, while heavy industry becomes especially consequential once clean supporters grow strong enough to trigger countermobilisation (Cheon and Urpelainen, 2013). The same logic appears at the firm level: low-adjustment-cost firms may support regulation that raises rivals' costs (Kennard, 2020), and low-adjustment-cost industries may accept climate goals in exchange for public support that de-risks transition and creates competitiveness opportunities (Kupzok and Nahm, 2024). Producer power therefore weakens through counterweight, defection, and acquiescence, not only through electoral pressure from voters.

The third lesson is that durability requires feedback. Activation can produce adoption, and adoption can produce emissions reductions in the short term. Durable stringency requires more. Enacted policy must change the future strategy space by altering producer strategies, voter experiences, institutional routines, or material constraints. Feedback can be supportive, negative, or stagnant. Supportive feedback makes continued stringency more attractive or less costly for later policymakers. Negative feedback makes backlash, dilution, or reversal more attractive. Stagnation occurs when policy persists formally while failing to reshape later politics.

Signal clarity determines how much time feedback has to mature. Under clear activation, policymakers face lower immediate reversal risk. Feedback can build gradually through investment, infrastructure, firm adjustment, consumer adoption, and institutional routines. Under noisy activation, the clock is shorter. Fast policy feedback must dampen opposition or build electorally relevant constituencies before opponents return, costs are reframed, or uncertainty chills investment. The

same instrument can therefore have different durability across configurations. A carbon price can survive when its incidence is buffered, its benefits are visible, and producer opposition is fragmented. A standard can fail when it creates visible household costs before supply chains and consumer benefits are ready. A subsidy can disappoint when it generates firm-level gains without voter attribution or electoral anchoring.

This reframes debates about instrument choice. The relevant question is less whether an instrument is a price, a standard, or a subsidy. The relevant question is what feedback profile it creates: who pays, who benefits, who observes, who attributes, who invests, and how quickly these effects become politically protective. Stokes's work is important here because it shows that policies expected to create new supportive politics can instead generate conflict during implementation (Stokes, 2020). The feedback mechanism is also fragile under uncertainty. Firms delay investment when future policy is unclear, and contested accumulation depends on precisely the investment feedback that uncertainty weakens (Gavriilidis et al., 2026). Credible and repeated signals can help move capital toward cleaner assets, especially during crises, but ambiguous signals do less (Bayer et al., 2025).

The fourth lesson is that sequencing works only with repeated activation. Early sequencing can create favourable feedback in tractable sectors, especially electricity and some forms of transport. Subsidies, standards, procurement, and infrastructure can build clean firms, consumer constituencies, and administrative capacity. These effects can make later stringency easier. Yet sequencing cannot carry the transition by itself. Feedback in one sector does not automatically spread to other sectors. Electricity feedback does not by itself create heat-pump supply chains, industrial hydrogen demand, agricultural methane policy, grid acceptance, or fossil-extraction phase-out coalitions.

Sequencing lowers the cost of the next political fight. It does not remove the need for that fight. Net zero requires repeated activation because policy must broaden into sectors where costs are more visible, losers are more concentrated, technologies may be less mature, and clean counterweights are weaker. Without renewed activation, early feedback can settle into partial transformation. The result is a durable climate framework in favourable sectors alongside stagnation elsewhere. This is why institutionalised ambition can coexist with implementation gaps. The framework therefore endorses sequencing as a route to durable stringency while rejecting the idea that sequencing alone can deliver net zero.

5.2 The future politics of net zero

The future politics of net zero follows from the same logic. Earlier successes were concentrated in sectors where cumulative transformation was easier. Electricity could be decarbonised through regulated utilities, renewable support, grid investment, contracts, and cost diffusion. Some transport transitions, especially electric vehicles in countries such as Norway, advanced through consumer benefits and infrastructure without directly threatening a powerful domestic producer coalition. These sectors mattered enormously. They also had unusually favourable feedback conditions.

Net zero now pushes climate politics into harder configurations. Heating, buildings, heavy industry, agriculture, land use, aviation, freight, local infrastructure, and fossil extraction expose the limits of earlier trajectories. These sectors have more visible household costs, more regionally concentrated losers, less mature supply chains, and stronger opportunities for opponents to frame policy as unfair, intrusive, or infeasible. They also pose sharper stranded-asset problems. Deep decarbonisation threatens fossil reserves, physical capital, carbon-complementary skills, local tax bases, and regional identities. As policy moves from clean-energy expansion to fossil-asset decline, dirty-sector cohesion may strengthen and salience may become more contested.

The framework therefore predicts more noisy contest in the next phase. Public support for climate goals remains important, but household-facing policies depend on beliefs about effectiveness, fairness, and personal cost. Recent evidence shows that support for climate policy is strongly shaped by perceived emissions effects, perceived effects on low-income households, and perceived effects on respondents' own households (Dechezleprêtre et al., 2025). Pocketbook effects matter especially for carbon taxes and other instruments with visible household costs (Beiser-McGrath and Bernauer, 2024). These findings reinforce the framework's emphasis on salience, signal clarity, cost visibility, and attribution.

The central task for future net-zero politics is consolidation plus extension. Consolidation protects feedback that already exists: clean producers, consumer constituencies, infrastructure, institutions, investment expectations, and administrative capacity. Extension creates feedback in sectors where cumulative transformation has not yet occurred. Consolidation without extension yields partial transformation. Extension without consolidation raises the risk of reversal. The political prob-

lem is therefore two-sided. Governments must defend existing feedback during periods of backlash while using new activation windows to broaden the transition into harder domains.

Institutions matter because they can support both tasks. Climate laws, advisory bodies, carbon budgets, ministries, and monitoring routines can keep climate on the agenda and make drift visible. Institutions that enable compensation, bargaining, and coordination can reduce the resistance of losers and help firms believe that transition bargains will hold. Yet institutions do not substitute for feedback in the sectors where emissions must fall. They shape attention, credibility, and bargaining conditions; sectoral transformation still requires producers, consumers, infrastructure, and voters to adjust. Future institutional reform should therefore be judged by whether it helps consolidate existing feedback and extend feedback into harder sectors.

Shocks and leadership also matter through the framework's variables. A shock is valuable for climate policy when it changes signal clarity, creates an anchor, lowers adjustment costs, fragments producers, or accelerates feedback. A natural disaster may make climate risk salient, yet its political effect depends on attribution and framing. An energy-price shock may produce cost backlash, or it may make fossil dependence salient as a security and competitiveness problem. A technology shock can lower adjustment costs and encourage producer fragmentation. Crisis politics helps climate policy when it changes the configuration, not simply when it raises attention.

Leadership matters where the office-seeking calculus has slack. Under latent demand with permissive cost conditions, policymakers can use low-salience instruments to create early feedback. Under noisy contest, they can act when an anchor exists and when policy can generate fast feedback. During shocks, they can choose frames, build anchors, invest in institutions, and send credible signals. Leadership has less scope where dirty producers remain unified, costs are visible, and activation is contested without an anchor. The practical lesson is clear: leadership works through configurations. It shapes salience, anchors, credibility, and feedback rather than escaping those constraints.

A final forward-looking issue concerns pressure valves such as carbon dioxide removal and international offsets. These tools may help governments preserve net-zero ambition when domestic cost confrontation becomes politically difficult. They may also weaken the domestic feedback on which durable stringency depends. If

future removals or foreign credits substitute for present domestic transformation, firms and households have fewer reasons to adjust today. The risk is therefore not only mitigation deterrence in the economic sense. It is feedback deterrence in the political-economy sense. Future net-zero policy will need to decide when such tools preserve ambition during adverse configurations and when they delay the feedback required for durable domestic stringency.

The politics ahead will therefore be less forgiving than the politics that delivered many early climate gains. Durable net-zero policy will require repeated activation, clearer signals, producer fragmentation, credible compensation, and feedback across sectors. The framework's central message is that governments cannot rely on targets, instruments, or public concern alone. They must use each favourable episode to change the future strategy space, while preparing for the noisy episodes that deeper decarbonisation will generate.

6 Conclusion

TBD

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